Md. Mehedi Hassan

mHealth Lab, Department of Biomedical Engineering, BUET, Dhaka
m.hassan.bme@gmail.com github.com/mubid-01 Portfolio

Highlights

Research Engineer at the mHealth Lab, Bangladesh University of Engineering and Technology, bridging machine learning and clinical diagnostics. My research develops robust, interpretable deep learning models for medical image analysis that integrate imperfect clinical data, including attention-gated models for fetal ultrasound, multiple-instance learning for Retinopathy of Prematurity screening, and hybrid networks for heterogeneous brain lesion segmentation.

Education

Bangladesh University of Engineering and Technology (BUET) *January 2022 - June 2026*
Dhaka, Bangladesh
Bachelor of Science in Biomedical Engineering (with Honours)

Relevant Coursework: Machine Learning for Biomedical Engineers, Medical Imaging, Digital Signal Processing, Biomedical Signals and Systems, Probability and Statistics, Linear Algebra, Bioinformatics, Engineering Optics for Medical Application

Notre Dame College *Year: 2020*
Dhaka, Bangladesh GPA: 5.00

Work Experience

Research Engineer, mHealth Lab *June 2026 – Present*
Bangladesh University of Engineering and Technology (BUET)

Collaborating on research projects with the Dept. of International Health, Johns Hopkins Bloomberg School of Public Health, USA, and the Institute of Global Health, University of Geneva, Switzerland.

Research Experiences

Undergraduate Thesis

Error-Resilient Fetal Ultrasound Dating and Adverse Pregnancy Outcome Prediction in Low-Resource Settings: A Multi-Modal Discrepancy Gating Framework *June 2025–Present*

Status: Manuscript under internal review before submission
Supervisor: Dr. Taufiq Hasan, Professor, Dept. of Biomedical Engineering, BUET
Principal Investigator: Dr. Abdullah Baqui, Professor, Dept. of International Health, Johns Hopkins University, USA

- Developed the "HeadsUp Attention Network" for **late-gestation fetal dating** and adverse pregnancy outcome (SGA, preterm birth, LBW) prediction on WHO AMANHI and Projahnmo Research Foundation cohorts, with a discrepancy-gating module that suppresses mistyped and missing clinical biometrics.
- Removed "Clever Hans" UI shortcuts via weakly supervised localization (**GA MAE 3.6–3.7 days**); offline tool in early clinical testing.

Ongoing Research Project

What Machine Learning Can and Cannot Do for Bedside Mpox Diagnosis: A Pre-specified Evaluation During a Clade Ib Outbreak in Eastern DR Congo *June 2026–Present*

Status: Manuscript in preparation
Supervisor: Dr. Taufiq Hasan, Professor, Dept. of Biomedical Engineering, BUET
Co-Supervisor: Dr. Andrew S. Azman, Institute of Global Health, University of Geneva, Switzerland

- Leading a **pre-specified, TRIPOD+AI-reported evaluation** of six ML model types for bedside **clade Ib mpox** diagnosis in 455 patients, with nested cross-validation and **temporal validation**.

- Despite **AUROC 0.70–0.71** (vs. 0.51 for the national case definition), no model met clinical-use criteria and **calibration failed to transfer**; extending to lesion-image and multimodal models.

Publications & Manuscripts

[1] **Md. Mehedi Hassan**, Taufiq Hasan. "Context-Aware Asymmetric Ensembling for Interpretable Retinopathy of Prematurity Screening via Active Query and Vascular Attention." ***Computers in Biology and Medicine*** **(Published, 2026)**. [10.1016/j.compbiomed.2026.111945]

- **Overview:** Dual-stream ensemble (MS-AQNet and VascuMIL) fusing clinical metadata with 4-channel imaging for ROP screening (0.99 AUC), with interpretable Vascular Threat Maps aligned to expert ophthalmological workflows.

[2] **Md. Mehedi Hassan**, Shafqat Alam, Shahriar Ahmed Seam, Maruf Ahmed. "SYNAPSE-Net: A Unified Framework with Lesion-Aware Hierarchical Gating for Robust Segmentation of Heterogeneous Brain Lesions." ***Computer Methods and Programs in Biomedicine Update*** **(Under Review, 2026)**. [arXiv:2510.26961]

- **Overview:** Hybrid CNN–Transformer framework for multi-modal MRI segmentation of heterogeneous brain lesions (WMH, ischemic stroke, glioblastoma) with lesion-aware hierarchical gating, evaluated on scanners absent from training.

[3] **Md. Mehedi Hassan**, Fariha Hasan Chowdhury, Sumaiya Islam Shotota, Nafisa Akter. "An Integrated Edge-AI Framework for Interpretable Respiratory Screening via Adaptive Cough Analysis in Low-Resource Settings" ***2nd International Conference on Advances in Computing, Communication, Electrical, and Smart Systems*** **(Accepted, 2026)**.

- **Overview:** Cough-screening pipeline with hysteresis gating and EfficientNet-B1 texture learning (85% patient-level accuracy), deployed as a fully offline Android app (Flutter–Chaquopy) for on-device screening.

Academic Projects

AeroIOP: A Low-Cost, Non-Contact Intraocular Pressure (IOP) Device | *Tools: SolidWorks, Arduino, Python*

- **Overview:** Portable non-contact tonometer with a calibrated air-puff mechanism, laser-photodetector and regression-based calibration, at ∼$180 versus $750+ for commercial devices.

Multi-Class Skin Cancer Classification Edge-AI App | *Tools: Python, TensorFlow, Streamlit, Flutter*

- **Overview:** Fine-tuned EfficientNet (B0–B4) on HAM10000 to classify 7 skin-lesion types from dermoscopic images (87.91% accuracy, 87% F1-score).

Test Scores

IELTS Academic: Overall **7.5** (Listening 7.5, Reading 8.0, Writing 7.5, Speaking 6.5)

Skills and Proficiencies

Programming	Python (PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy), C/C++, MATLAB, R
Computing Skills	CNNs, RNNs, Transformers, LLMs; supervised and unsupervised learning
Image Processing Tools	MATLAB, ITKSnap, BET-FSL, CVAT, VoTT
Software & Tools	Git, LaTeX, Ansys Fluent, COMSOL, SolidWorks, MS Office
Other	Canva, Adobe Illustrator; Windows, Linux

Achievements

Dean's List, Faculty of Electrical and Electronic Engineering — *Multiple Times*
Champion, BME Design Day Competition, BUET — *2023*
Second Runners-up, Poster Presentation, ESONANCE, IUT — *2023*

Training & Voluntary Works

Training, National Electro-Medical Equipment Maintenance Workshop, Ministry of Health — *March 2025*

Creative Lead, BUET Biomedical Engineering Society (BMES) Student Chapter — *August 2024 - Present*

Enzyme, Bangladesh Biology Olympiad — *2022*